

Microring Resonators as analog processors - Introduction paper

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1 MRR as first-order linear ODE solver

We began our work by demonstrating that a microring resonator (MRR) can solve first-order linear ordinary differential equations. We analyzed the impact of the input signal bandwidth by computing the cavity lifetime of the proposed architecture and illustrating the system’s behaviour under different bandwidth conditions.

After selecting a suitable input bandwidth, we demonstrated that a MRR without a tunable k parameter, can solve — with a low RMSE — only a single specific equation.

To address this limitation, we introduced the approach proposed in the paper *”All-optical differential equation solver with constant-coefficient tunable based on a single microring resonator. — Yang et al.”*. The authors demonstrate that the microring’s k coefficient can be tuned via voltage injection through a p-i-n junction. This changes the refractive index and absorption coefficient (*Electrooptical Effects in Silicon. — Soref, Bennet*), resulting in a change in the quality factor, and therefore on the k coefficient.

Following the tuning of the k -parameter, this leads to a significant improvement of the performance in the covered area. Since the paper does not demonstrate if the tuning operation also works for other values of voltage, we assumed that the MRR is tuned automatically with a feasible value of k that minimizes the RMSE.

Concluding the first part, we evaluated the impact of the phase detuning phenomenon. By simulating different phase offsets, we demonstrated how phase misalignment affects both the device’s power losses and the overall RMSE.

2 Cascading MRR to solve N-th order linear ODE

Building on the tunable single-ring solver, we extended the architecture to solve higher-order linear differential equations by cascading N microring resonators in series.

Since the drop-port response of each ring behaves as a first-order transfer function of the form $1/(j\omega + k)$, connecting N rings in a chain multiplies their individual responses, yielding an overall transfer function $\prod_{i=1}^N 1/(j\omega + k_i)$. This product corresponds to the operator $\prod_{i=1}^N (d/dt + k_i) y(t) = x(t)$, so a cascade of N rings solves an N -th order linear ODE.

We validated the cascade against the direct solution of the target equation, and characterized its power loss and RMSE as the order increases, for both Gaussian and super-Gaussian inputs.

3 MRR as Linear Time-Invariant differential equation solver

In the final section of this work, we introduced a novel microring resonator (*”Compact tunable silicon photonic differential-equation solver for general linear time-invariant systems. — Wu et al.”*) architecture specifically engineered to address and solve Linear Time-Invariant (LTI) differential equations within the optical domain.

To validate the theoretical framework, the proposed MRR configuration was modeled and implemented using the MATLAB environment.

Through simulations, we demonstrated the system’s dynamic reconfigurability, highlighting how thermal tuning via integrated microheaters allows for parameter adjustments.

Furthermore, we expanded the scope of this architecture by introducing a scalable extension capable of solving second-order differential equations.